MATH 128a: Homework #10

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Due by: **Wednesday, November 12, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 3.4

Solve problems 2a, 4a, 11 of Section 3.4 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 1

Problem 2
Part (a) Given data:

x	$f(x)$	$f'(x)$
0	1.00000	2.00000
0.5	2.71828	5.43656

Using Theorem 3.9 (Hermite interpolation), we construct $H_3(x)$, a polynomial of degree at most 3. The Hermite polynomial has the form:

$$H_3(x) = \sum_{j=0}^{1} f(x_j)H_j(x) + \sum_{j=0}^{1} f'(x_j)\hat{H}_j(x)$$

where $H_j(x) = [1 - 2(x - x_j)L_j'(x_j)]L_j^2(x)$ and $\hat{H}_j(x) = (x - x_j)L_j^2(x)$.
For two points, $L_0(x) = \frac{x-x_1}{x_0-x_1} = \frac{x-0.5}{0-0.5} = 2(0.5 - x)$ and $L_1(x) = \frac{x-x_0}{x_1-x_0} = \frac{x-0}{0.5-0} = 2x$.
Computing derivatives: $L_0'(x) = -2$ and $L_1'(x) = 2$.
At $x_0 = 0$: $L_0'(0) = -2$, so $H_0(x) = [1 - 2(x - 0)(-2)][2(0.5 - x)]^2 = (1 + 4x)(0.5 - x)^2 \cdot 4$
At $x_1 = 0.5$: $L_1'(0.5) = 2$, so $H_1(x) = [1 - 2(x - 0.5)(2)](2x)^2 = [1 - 4(x - 0.5)] \cdot 4x^2$
$\hat{H}_0(x) = (x - 0)[2(0.5 - x)]^2 = 4x(0.5 - x)^2$
$\hat{H}_1(x) = (x - 0.5)(2x)^2 = 4x^2(x - 0.5)$
Therefore:

$$\begin{aligned} H_3(x) = {} & 1.00000 \cdot (1 + 4x) \cdot 4(0.5 - x)^2 + 2.71828 \cdot [1 - 4(x - 0.5)] \cdot 4x^2 \\ & + 2.00000 \cdot 4x(0.5 - x)^2 + 5.43656 \cdot 4x^2(x - 0.5) \end{aligned}$$

Expanding and simplifying:

$$\boxed{H_3(x) = 1 + 2x + 2x^2 + 1.71828x^3}$$

Problem 4 Part (a) Given: $f(x) = e^{2x}$ and we need to approximate $f(0.43)$ using the polynomial from Problem 2(a).
From Problem 2(a), we have:

$$H_3(x) = 1 + 2x + 2x^2 + 1.71828x^3$$

Evaluating at $x = 0.43$:

$$\begin{aligned} H_3(0.43) &= 1 + 2(0.43) + 2(0.43)^2 + 1.71828(0.43)^3 \\ &= 1 + 0.86 + 2(0.1849) + 1.71828(0.079507) \\ &= 1 + 0.86 + 0.3698 + 0.13664 \\ &= 2.36644 \end{aligned}$$

The true value is:

$$f(0.43) = e^{2(0.43)} = e^{0.86} = 2.36316$$

The absolute error is:

$$|f(0.43) - H_3(0.43)| = |2.36316 - 2.36644| = \boxed{0.00328}$$

Problem 11
Part (a): Uniqueness of $H_{2n+1}(x)$
Theorem: $H_{2n+1}(x)$ is the unique polynomial of degree at most $2n + 1$ such that

$$H_{2n+1}(x_i) = f(x_i) \quad \text{and} \quad H'_{2n+1}(x_i) = f'(x_i)$$

for $i = 0, 1, \ldots, n$.
Proof:
Assume $P(x)$ is another polynomial of degree at most $2n+1$ satisfying the same conditions. Define:

$$D(x) = H_{2n+1}(x) - P(x)$$

Then $D(x)$ is a polynomial of degree at most $2n + 1$, and:

$$\begin{aligned} D(x_i) &= H_{2n+1}(x_i) - P(x_i) = f(x_i) - f(x_i) = 0 \\ D'(x_i) &= H'_{2n+1}(x_i) - P'(x_i) = f'(x_i) - f'(x_i) = 0 \end{aligned}$$

for $i = 0, 1, \ldots, n$.
This means that $D(x)$ has zeros at $x_0, x_1, \ldots, x_n$, and $D'(x)$ also has zeros at these same points. If $D'(x_i) = 0$, then x_i is a critical point of $D(x)$. Since $D(x_i) = 0$ as well, x_i is a zero of multiplicity at least 2 for $D(x)$.
Therefore, $D(x)$ can be written as:

$$D(x) = (x - x_0)^2(x - x_1)^2 \cdots (x - x_n)^2 Q(x)$$

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for some polynomial $Q(x)$.
The degree of $(x-x_0)^2(x-x_1)^2\cdots(x-x_n)^2$ is $2(n+1)=2n+2$.
Since $\deg(D)\le 2n+1<2n+2$, we must have $Q(x)=0$.
Therefore, $D(x)=0$, which implies $H_{2n+1}(x)=P(x)$.
Thus, $H_{2n+1}(x)$ is unique. Q.E.D.

Part (b): Derive the Error Term in Theorem 3.9

Following the hint, define:

$$g(t)=f(t)-H_{2n+1}(t)-\frac{(t-x_0)^2\cdots(t-x_n)^2}{(x-x_0)^2\cdots(x-x_n)^2}[f(x)-H_{2n+1}(x)]$$

where x is a fixed point in $[a,b]$ (not equal to any x_i).

Observations:

1. $g(x_i)=f(x_i)-H_{2n+1}(x_i)-0=0$ for $i=0,1,\ldots,n$
2. $g(x)=f(x)-H_{2n+1}(x)-[f(x)-H_{2n+1}(x)]=0$
3. $g'(x_i)=f'(x_i)-H'_{2n+1}(x_i)-0=0$ for $i=0,1,\ldots,n$

So $g(t)$ has zeros at $x_0,x_1,\ldots,x_n,x$ (that's $n+2$ zeros), and $g'(t)$ has zeros at $x_0,x_1,\ldots,x_n$ (that's $n+1$ zeros).
By Rolle's theorem, between consecutive zeros of g, there exists at least one zero of g'. So g' has at least $n+1$ additional zeros beyond $x_0,\ldots,x_n$, giving a total of at least $2n+2$ zeros for g'.
Continuing this process:

- $g''(t)$ has at least $2n+1$ zeros
- $g'''(t)$ has at least $2n$ zeros
- $\vdots$
- $g^{(2n+2)}(t)$ has at least 1 zero in $[a,b]$

Let $\xi\in(a,b)$ be a zero of $g^{(2n+2)}(t)$. Then:

$$g^{(2n+2)}(\xi)=0$$

Now compute:

$$g^{(2n+2)}(t)=f^{(2n+2)}(t)-H_{2n+1}^{(2n+2)}(t)-\frac{(2n+2)!}{(x-x_0)^2\cdots(x-x_n)^2}[f(x)-H_{2n+1}(x)]$$

Since $H_{2n+1}(t)$ has degree at most $2n+1$, we have $H_{2n+1}^{(2n+2)}(t)=0$.
Therefore:

$$0=f^{(2n+2)}(\xi)-\frac{(2n+2)!}{(x-x_0)^2\cdots(x-x_n)^2}[f(x)-H_{2n+1}(x)]$$

Solving for the error:

$$f(x)-H_{2n+1}(x)=\frac{(x-x_0)^2\cdots(x-x_n)^2}{(2n+2)!}f^{(2n+2)}(\xi)$$

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Therefore, the error term is:

$$\boxed{E(x) = \frac{(x-x_0)^2(x-x_1)^2\cdots(x-x_n)^2}{(2n+2)!} f^{(2n+2)}(\xi)}$$

for some $\xi \in (a,b)$ (or in the smallest interval containing $x, x_0, x_1, \ldots, x_n$).

Problem 2: Problems from the textbook, chapter 4.1

Solve problems 2a, 4a, 5d, 24(22) of Section 4.1 of the textbook.

In 24(22), use a linear combination of $f(x_0-h)$, $f(x_0)$, $f(x_0+h)$, $f(x_0+2h)$ and $f(x_0+3h)$ (i.e., also use $f(x_0)$ contrary to what the hint suggests)
(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 2

Problem 2(a)
Forward Difference Formula (for the first point): The forward difference formula for $f'(x)$ is:

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0+h) - f(x_0+2h)}{2h}$$

For $x_0 = -0.3$, $x_1 = -0.1$, $x_2 = 0.1$:

$$\begin{aligned} f'(-0.3) &\approx \frac{-3f(-0.3) + 4f(-0.1) - f(0.1)}{2(0.2)} \\ &= \frac{-3(1.9507) + 4(2.0421) - 2.0601}{0.4} \\ &= \frac{-5.8521 + 8.1684 - 2.0601}{0.4} \\ &= \frac{0.2562}{0.4} \\ &= \boxed{0.6405} \end{aligned}$$

Central Difference Formula (for the middle point): The central difference formula is:

$$f'(x_1) \approx \frac{f(x_2) - f(x_0)}{2h}$$

For $x_1 = -0.1$:

$$\begin{aligned} f'(-0.1) &\approx \frac{f(0.1) - f(-0.3)}{2(0.2)} \\ &= \frac{2.0601 - 1.9507}{0.4} \\ &= \frac{0.1094}{0.4} \\ &= \boxed{0.2735} \end{aligned}$$

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Backward Difference Formula (for the last point): The backward difference formula is:

$$f'(x_2) \approx \frac{3f(x_2) - 4f(x_1) + f(x_0)}{2h}$$

For $x_2 = 0.1$:

$$\begin{aligned} f'(0.1) &\approx \frac{3f(0.1) - 4f(-0.1) + f(-0.3)}{2(0.2)} \\ &= \frac{3(2.0601) - 4(2.0421) + 1.9507}{0.4} \\ &= \frac{6.1803 - 8.1684 + 1.9507}{0.4} \\ &= \frac{-0.0374}{0.4} \\ &= \boxed{-0.0935} \end{aligned}$$

Problem 4(a)
Given: $f(x) = 2\cos(2x) - x$
First, compute the exact derivatives at each point:

$$f'(x) = -4\sin(2x) - 1$$

$$f''(x) = -8\cos(2x)$$

$$f'''(x) = 16\sin(2x)$$

At $x = -0.3$: Exact derivative:

$$f'(-0.3) = -4\sin(-0.6) - 1 = -4(-0.5646) - 1 = 2.2584 - 1 = 1.2584$$

Approximate derivative (from Problem 2a): $f'(-0.3) \approx 0.6405$
Actual error:

$$|f'(-0.3) - 0.6405| = |1.2584 - 0.6405| = \boxed{0.6179}$$

Error bound for three-point forward difference formula:

$$|E| \leq \frac{h^2}{3} \max_{x \in [-0.3, 0.1]} |f'''(x)|$$

Find maximum of $|f'''(x)| = |16\sin(2x)|$ on $[-0.3, 0.1]$: - At $x = -0.3$: $|16\sin(-0.6)| = |16(-0.5646)| = 9.034$ - At $x = 0.1$: $|16\sin(0.2)| = |16(0.1987)| = 3.179$ - Maximum occurs at $x = -0.3$: $\max |f'''(x)| \approx 9.034$
Error bound:

$$|E| \leq \frac{(0.2)^2}{3}(9.034) = \frac{0.04}{3}(9.034) = \boxed{0.1205}$$

At $x = -0.1$: Exact derivative:

$$f'(-0.1) = -4\sin(-0.2) - 1 = -4(-0.1987) - 1 = 0.7948 - 1 = -0.2052$$

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Approximate derivative: $f'(-0.1) \approx 0.2735$
Actual error:

$$|f'(-0.1) - 0.2735| = |-0.2052 - 0.2735| = \boxed{0.4787}$$

Error bound for central difference formula:

$$|E| \leq \frac{h^2}{6} \max_{x \in [-0.3, 0.1]} |f'''(x)|$$

$$|E| \leq \frac{(0.2)^2}{6}(9.034) = \frac{0.04}{6}(9.034) = \boxed{0.0602}$$

At $x = 0.1$: Exact derivative:

$$f'(0.1) = -4\sin(0.2) - 1 = -4(0.1987) - 1 = -0.7948 - 1 = -1.7948$$

Approximate derivative: $f'(0.1) \approx -0.0935$
Actual error:

$$|f'(0.1) - (-0.0935)| = |-1.7948 + 0.0935| = \boxed{1.7013}$$

Error bound for three-point backward difference formula:

$$|E| \leq \frac{h^2}{3} \max_{x \in [-0.3, 0.1]} |f'''(x)|$$

$$|E| \leq \frac{(0.2)^2}{3}(9.034) = \boxed{0.1205}$$

Problem 5(d):
At $x = 2.0$ (use forward difference):

$$f'(2.0) \approx \frac{-3f(2.0) + 4f(2.1) - f(2.2)}{2h}$$

$$\begin{aligned} f'(2.0) &= \frac{-3(3.6887983) + 4(3.6905701) - 3.6688192}{2(0.1)} \\ &= \frac{-11.0663949 + 14.7622804 - 3.6688192}{0.2} \\ &= \frac{0.0270663}{0.2} \\ &= \boxed{0.1353} \end{aligned}$$

At $x = 2.1$ (use central difference):

$$f'(2.1) \approx \frac{f(2.2) - f(2.0)}{2h}$$

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$$\begin{aligned} f'(2.1) &= \frac{3.6688192 - 3.6887983}{2(0.1)} \\ &= \frac{-0.0199791}{0.2} \\ &= \boxed{-0.0999} \end{aligned}$$

At $x = 2.2$ (use central difference):

$$f'(2.2) \approx \frac{f(2.3) - f(2.1)}{2h}$$

$$\begin{aligned} f'(2.2) &= \frac{3.6245909 - 3.6905701}{2(0.1)} \\ &= \frac{-0.0659792}{0.2} \\ &= \boxed{-0.3299} \end{aligned}$$

At $x = 2.3$ (use backward difference):

$$f'(2.3) \approx \frac{3f(2.3) - 4f(2.2) + f(2.1)}{2h}$$

$$\begin{aligned} f'(2.3) &= \frac{3(3.6245909) - 4(3.6688192) + 3.6905701}{2(0.1)} \\ &= \frac{10.8737727 - 14.6752768 + 3.6905701}{0.2} \\ &= \frac{-0.1109340}{0.2} \\ &= \boxed{-0.5547} \end{aligned}$$

Problem 24:
Step 1: Set up the linear combination:
Consider the expression:

$$f'(x_0) \approx Af(x_0 - h) + Bf(x_0) + Cf(x_0 + h) + Df(x_0 + 2h) + Ef(x_0 + 3h)$$

Step 2: Taylor series expansions:
Expand each function value in a Taylor series about x_0:

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$$
\begin{aligned}
f(x_0) &= f(x_0) \\
f(x_0+h) &= f(x_0) + hf'(x_0) + \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(x_0) + \frac{h^4}{24}f^{(4)}(x_0) + O(h^5) \\
f(x_0-h) &= f(x_0) - hf'(x_0) + \frac{h^2}{2}f''(x_0) - \frac{h^3}{6}f'''(x_0) + \frac{h^4}{24}f^{(4)}(x_0) + O(h^5) \\
f(x_0+2h) &= f(x_0) + 2hf'(x_0) + \frac{4h^2}{2}f''(x_0) + \frac{8h^3}{6}f'''(x_0) + \frac{16h^4}{24}f^{(4)}(x_0) + O(h^5) \\
f(x_0+3h) &= f(x_0) + 3hf'(x_0) + \frac{9h^2}{2}f''(x_0) + \frac{27h^3}{6}f'''(x_0) + \frac{81h^4}{24}f^{(4)}(x_0) + O(h^5)
\end{aligned}
$$

Step 3: Substitute into the linear combination:

$$
\begin{aligned}
&Af(x_0-h) + Bf(x_0) + Cf(x_0+h) + Df(x_0+2h) + Ef(x_0+3h) \\
&= (A+B+C+D+E)f(x_0) \\
&\quad + h(-A+C+2D+3E)f'(x_0) \\
&\quad + \frac{h^2}{2}(A+C+4D+9E)f''(x_0) \\
&\quad + \frac{h^3}{6}(-A+C+8D+27E)f'''(x_0) \\
&\quad + \frac{h^4}{24}(A+C+16D+81E)f^{(4)}(x_0) \\
&\quad + O(h^5)
\end{aligned}
$$

Step 4: Set up the system of equations:
For the approximation to equal $f'(x_0)$ with error $O(h^4)$, we need:

$$
\begin{aligned}
A+B+C+D+E &= 0 \quad \text{(coefficient of } f(x_0)\text{)} && (1)\\
-A+C+2D+3E &= 1 \quad \text{(coefficient of } hf'(x_0)\text{)} && (2)\\
A+C+4D+9E &= 0 \quad \text{(coefficient of } h^2f''(x_0)\text{)} && (3)\\
-A+C+8D+27E &= 0 \quad \text{(coefficient of } h^3f'''(x_0)\text{)} && (4)\\
A+C+16D+81E &= 0 \quad \text{(coefficient of } h^4f^{(4)}(x_0)\text{)} && (5)
\end{aligned}
$$

Step 5: Solve the system:
From equations (3) and (5):

$$
\begin{aligned}
A+C+4D+9E &= 0 \quad ...(3)\\
A+C+16D+81E &= 0 \quad ...(5)
\end{aligned}
$$

Subtracting (3) from (5):

$$12D + 72E = 0 \implies D = -6E$$

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From equations (3) and (4):

$$A + C + 4D + 9E = 0 \quad ...(3)$$
$$-A + C + 8D + 27E = 0 \quad ...(4)$$

Adding (3) and (4):

$$2C + 12D + 36E = 0$$

Substituting $D = -6E$:

$$2C + 12(-6E) + 36E = 0$$
$$2C - 72E + 36E = 0$$
$$2C - 36E = 0$$
$$C = 18E$$

From equation (3):

$$A + 18E + 4(-6E) + 9E = 0$$
$$A + 18E - 24E + 9E = 0$$
$$A + 3E = 0$$
$$A = -3E$$

From equation (2):

$$-(-3E) + 18E + 2(-6E) + 3E = 1$$
$$3E + 18E - 12E + 3E = 1$$
$$12E = 1$$
$$E = \frac{1}{12}$$

Therefore:

$$E = \frac{1}{12}$$
$$D = -6E = -\frac{6}{12} = -\frac{1}{2}$$
$$C = 18E = \frac{18}{12} = \frac{3}{2}$$
$$A = -3E = -\frac{3}{12} = -\frac{1}{4}$$

From equation (1):

$$B = -A - C - D - E = \frac{1}{4} - \frac{3}{2} + \frac{1}{2} - \frac{1}{12}$$
$$B = \frac{3 - 18 + 6 - 1}{12} = \frac{-10}{12} = -\frac{5}{6}$$

Final Formula:

$$f'(x_0) = \frac{1}{12h}\left[-3f(x_0 - h) - 10f(x_0) + 18f(x_0 + h) - 6f(x_0 + 2h) + f(x_0 + 3h)\right] + O(h^4)$$

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Problem 3: Problems from the textbook, chapter 4.2

Solve problems 1a, 5, 9 10 of Section 4.2 of the textbook.

In 5: use the notation of Lecture 19, i.e. assume $N_2(h/2^j)$ has the given values, that $M = N_2(h) + K_2h^2 + K_4h^4 + K_6h^6 + K_8h^8 + O(h^{10})$, and determine $N_8(h)$.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 3

Problem 1(a):
Given: $f(x) = \ln x$, $x_0 = 1.0$, $h = 0.4$
We want to find $N_3(h)$, which means we need three levels of extrapolation.
Computing $N_1(h)$ values using centered difference formula
The centered difference formula is:

$$N_1(h) = \frac{f(x_0 + h) - f(x_0 - h)}{2h}$$

For $h = 0.4$:

$$N_1(0.4) = \frac{\ln(1.4) - \ln(0.6)}{2(0.4)} = \frac{0.336472 - (-0.510826)}{0.8}$$
$$= \frac{0.847298}{0.8} = 1.059123$$

For $h = 0.2$:

$$N_1(0.2) = \frac{\ln(1.2) - \ln(0.8)}{2(0.2)} = \frac{0.182322 - (-0.223144)}{0.4}$$
$$= \frac{0.405465}{0.4} = 1.013663$$

For $h = 0.1$:

$$N_1(0.1) = \frac{\ln(1.1) - \ln(0.9)}{2(0.1)} = \frac{0.095310 - (-0.105361)}{0.2}$$
$$= \frac{0.200671}{0.2} = 1.003355$$

For $h = 0.05$:

$$N_1(0.05) = \frac{\ln(1.05) - \ln(0.95)}{2(0.05)} = \frac{0.048790 - (-0.051293)}{0.1}$$
$$= \frac{0.100083}{0.1} = 1.000833$$

Richardson Extrapolation Table
The extrapolation formula is:

$$N_{j+1}(h) = N_j(h) + \frac{N_j(h) - N_j(2h)}{4^j - 1}$$

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h	$N_1(h)$	$N_2(h)$	$N_3(h)$	$N_4(h)$
0.4	1.059123			
0.2	1.013663	0.998510		
0.1	1.003355	0.999918	1.000012	
0.05	1.000833	0.999998	1.000000	1.000000

Calculations:
$N_2(0.2)$:

$$N_2(0.2) = N_1(0.2) + \frac{N_1(0.2) - N_1(0.4)}{4^1 - 1} = 1.013663 + \frac{1.013663 - 1.059123}{3} = 1.013663 - 0.015153 = 0.998510$$

$N_2(0.1)$:

$$N_2(0.1) = 1.003355 + \frac{1.003355 - 1.013663}{3} = 1.003355 - 0.003436 = 0.999918$$

$N_2(0.05)$:

$$N_2(0.05) = 1.000833 + \frac{1.000833 - 1.003355}{3} = 1.000833 - 0.000841 = 0.999998$$

$N_3(0.1)$:

$$N_3(0.1) = N_2(0.1) + \frac{N_2(0.1) - N_2(0.2)}{4^2 - 1} = 0.999918 + \frac{0.999918 - 0.998510}{15} = 0.999918 + 0.000094 = 1.000012$$

$N_3(0.05)$:

$$N_3(0.05) = 0.999998 + \frac{0.999998 - 0.999918}{15} = 0.999998 + 0.000005 = 1.000003$$

Therefore:

$$\boxed{N_3(0.4) \approx 1.000012}$$

Note: The exact value is $f'(1) = \frac{1}{x}\Big|_{x=1} = 1$.

Problem 5
Given data for $M = \int_0^\pi \sin x \, dx$:

$$N_1(h) = 1.570796$$
$$N_1(h/2) = 1.896119$$
$$N_1(h/4) = 1.974232$$
$$N_1(h/8) = 1.993570$$

Given: $M = N_1(h) + K_1 h^2 + K_2 h^4 + K_3 h^6 + K_4 h^8 + O(h^{10})$
Following the hint, we use $N_2(h/2^j)$ notation where the values are at step sizes $h/2^j$.
The extrapolation formula for even powers is:

$$N_{j+1}(h) = N_j(h) + \frac{N_j(h) - N_j(2h)}{4^j - 1}$$

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Extrapolation Table

Step	$N_2(h/2^j)$	$N_4(h/2^j)$	$N_6(h/2^j)$	$N_8(h/2^j)$
h	1.570796			
$h/2$	1.896119	2.004536		
$h/4$	1.974232	2.000269	1.999984	
$h/8$	1.993570	2.000017	1.999999	2.000000

Calculations:
$N_4(h/2)$ (eliminating h^2 term):

$$N_4(h/2) = N_2(h/2) + \frac{N_2(h/2) - N_2(h)}{4^1 - 1} = 1.896119 + \frac{1.896119 - 1.570796}{3}$$

$$= 1.896119 + \frac{0.325323}{3} = 1.896119 + 0.108441 = 2.004560$$

$N_4(h/4)$:

$$N_4(h/4) = 1.974232 + \frac{1.974232 - 1.896119}{3} = 1.974232 + 0.026038 = 2.000270$$

$N_4(h/8)$:

$$N_4(h/8) = 1.993570 + \frac{1.993570 - 1.974232}{3} = 1.993570 + 0.006446 = 2.000016$$

$N_6(h/4)$ (eliminating h^4 term):

$$N_6(h/4) = N_4(h/4) + \frac{N_4(h/4) - N_4(h/2)}{4^2 - 1} = 2.000270 + \frac{2.000270 - 2.004560}{15}$$

$$= 2.000270 - 0.000286 = 1.999984$$

$N_6(h/8)$:

$$N_6(h/8) = 2.000016 + \frac{2.000016 - 2.000270}{15} = 2.000016 - 0.000017 = 1.999999$$

$N_8(h/8)$ (eliminating h^6 term):

$$N_8(h/8) = N_6(h/8) + \frac{N_6(h/8) - N_6(h/4)}{4^3 - 1} = 1.999999 + \frac{1.999999 - 1.999984}{63}$$

$$= 1.999999 + 0.000000238 = 1.999999$$

Therefore:

$$\boxed{N_8(h) \approx 2.000000}$$

The exact value is $M = \int_0^\pi \sin x\, dx = [-\cos x]_0^\pi = -\cos\pi + \cos 0 = 1 + 1 = 2$.

Problem 9
Given: $M = N(h) + K_1 h + K_2 h^2 + K_3 h^3 + \cdots$
We have $N(h)$, $N(h/3)$, and $N(h/6)$.

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First Extrapolation (eliminate h term)
To eliminate the $K_1 h$ term, we use:

$$N_1(h) = \frac{3N(h/3) - N(h)}{3-1} = \frac{3N(h/3) - N(h)}{2}$$

For step size $h/3$:

$$N_1(h/3) = \frac{3N(h/6) - N(h/3)}{2}$$

After eliminating the h term, we have:

$$M = N_1(h) + K_2' h^2 + K_3' h^3 + \cdots$$

Second Extrapolation (eliminate h^2 term)
Now eliminate the h^2 term:

$$N_2(h) = \frac{9N_1(h/3) - N_1(h)}{9-1} = \frac{9N_1(h/3) - N_1(h)}{8}$$

After this step, we have:

$$M = N_2(h) + K_3'' h^3 + O(h^4)$$

Final Formula

$$\boxed{M \approx N_2(h) = \frac{9N_1(h/3) - N_1(h)}{8}}$$

where

$$N_1(h/3) = \frac{3N(h/6) - N(h/3)}{2}, \quad N_1(h) = \frac{3N(h/3) - N(h)}{2}$$

Substituting:

$$M \approx \frac{9 \cdot \frac{3N(h/6) - N(h/3)}{2} - \frac{3N(h/3) - N(h)}{2}}{8}$$

$$= \frac{27N(h/6) - 9N(h/3) - 3N(h/3) + N(h)}{16}$$

$$\boxed{M \approx \frac{27N(h/6) - 12N(h/3) + N(h)}{16} + O(h^3)}$$

Problem 10:
Given: $M = N(h) + K_1 h^2 + K_2 h^4 + K_3 h^6 + \cdots$
We have $N(h)$, $N(h/3)$, and $N(h/6)$.
First Extrapolation (eliminate h^2 term)
To eliminate the $K_1 h^2$ term:

$$N_1(h) = \frac{9N(h/3) - N(h)}{9-1} = \frac{9N(h/3) - N(h)}{8}$$

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For step size $h/3$:

$$N_1(h/3) = \frac{9N(h/6) - N(h/3)}{8}$$

After eliminating the h^2 term:

$$M = N_1(h) + K_2' h^4 + K_3' h^6 + \cdots$$

Second Extrapolation (eliminate h^4 term)
Now eliminate the h^4 term using ratio $3^4 = 81$:

$$N_2(h) = \frac{81N_1(h/3) - N_1(h)}{81 - 1} = \frac{81N_1(h/3) - N_1(h)}{80}$$

After this step:

$$M = N_2(h) + K_3'' h^6 + O(h^8)$$

Final Formula

$$\boxed{M \approx N_2(h) = \frac{81N_1(h/3) - N_1(h)}{80}}$$

where

$$N_1(h/3) = \frac{9N(h/6) - N(h/3)}{8}, \quad N_1(h) = \frac{9N(h/3) - N(h)}{8}$$

Substituting:

$$M \approx \frac{81 \cdot \frac{9N(h/6) - N(h/3)}{8} - \frac{9N(h/3) - N(h)}{8}}{80}$$

$$= \frac{729N(h/6) - 81N(h/3) - 9N(h/3) + N(h)}{640}$$

$$\boxed{M \approx \frac{729N(h/6) - 90N(h/3) + N(h)}{640} + O(h^6)}$$

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Problem 4: Derivation of finite difference formula

Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth, bounded function and $\bar{x} \in \mathbb{R}$. A one-sided difference approximation to $f'(\bar{x})$ based on $f(\bar{x})$, $f(\bar{x} - h)$ and $f(\bar{x} - 2h)$ is given by

$$D_2 f(\bar{x}) = \frac{3f(\bar{x}) - 4f(\bar{x} - h) + f(\bar{x} - 2h)}{2h}. \tag{6}$$

Determine the quadratic polynomial $p(x)$ that interpolates f at the points $\bar{x}$, $\bar{x} - h$ and $\bar{x} - 2h$. Verify that evaluation $p'(\bar{x})$ gives equation (6).

Answer of Problem 4

Using the Newton polynomial, p is given by

$$p(x) = c_0 + c_1(x - \bar{x}) + c_2(x - \bar{x})(x - (\bar{x} - h)),$$

for some coefficients c_0, c_1 and c_2. Using the conditions $p(\bar{x}) = u(\bar{x})$, $p(\bar{x} - h) = u(\bar{x} - h)$ and $p(\bar{x} - 2h) = u(\bar{x} - 2h)$, we can determine these as

$$p(\bar{x}) = c_0 = u(\bar{x}),$$

then inserting this and solving for c_1:

$$\begin{aligned} p(\bar{x} - h) = c_0 - hc_1 = u(\bar{x}) - hc_1 &\overset{!}{=} u(\bar{x} - h) \\ \Rightarrow \quad c_1 &= \frac{u(\bar{x}) - u(\bar{x} - h)}{h}, \end{aligned}$$

and then inserting c_0 and c_1 in the remaining equation, we obtain:

$$\begin{aligned} p(\bar{x} - 2h) = c_0 - 2hc_1 + 2h^2 c_2 = u(\bar{x}) - 2(u(\bar{x}) - u(\bar{x} - h)) + 2h^2 c_2 &\overset{!}{=} u(\bar{x} - 2h) \\ \Rightarrow \quad c_2 &= \frac{u(\bar{x} - 2h) + u(\bar{x}) - 2u(\bar{x} - h)}{2h^2}. \end{aligned}$$

The derivative of p is given by

$$p'(x) = c_1 + c_2(2x - 2\bar{x} + h).$$

Inserting the above expressions for c_1 and c_2 and evaluating at $\bar{x}$, we get

$$p'(\bar{x}) = c_1 + hc_2 = \frac{3u(\bar{x}) - 4u(\bar{x} - h) + u(\bar{x} - 2h)}{2h}$$

which is exactly (6).

Problem 5: Use of `fdstencil`

a) Use the method from Lecture 19 with the Taylor expansion to set up the 5×5 linear system that would determine a fourth-order accurate finite difference approximation to $f''(x)$ based on 5 equally spaced points,

$$f''(x) = c_{-2} f(x - 2h) + c_{-1} f(x - h) + c_0 f(x) + c_1 f(x + h) + c_2 f(x + 2h) + O(h^4).$$

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b) Compute the coefficients using the MATLAB code `fdstencil.m` available from the website, and check that they satisfy the system you determined in part (a).

c) Test this finite difference formula to approximate $f''(1)$ for $f(x) = \sin(2x)$ with values of h from the array `hvals = logspace(-1, -4, 13)`. Make a table of the error vs. h for several values of h and compare against the predicted error from the leading term of the expression printed by `fdstencil`. guidance on how to make such a table.

Also produce a log-log plot of the absolute value of the error vs. h.

You should observe the predicted accuracy for larger values of h. For smaller values, numerical cancellation in computing the linear combination of u values impacts the accuracy observed.

Answer of Problem 5

1. Proceeding as in the lecture or using equation (1.19) in Chapter 1 of the book 'Finite Difference Methods for Ordinary and Partial Differential Equations' by R.J. LeVeque, we obtain the Vandermonde matrix

$$\mathbf{V} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2h & -h & 0 & h & 2h \\ 2h^2 & \frac{h^2}{2} & 0 & \frac{h^2}{2} & 2h^2 \\ -\frac{4}{3}h^3 & -\frac{h^3}{6} & 0 & \frac{h^3}{6} & \frac{4}{3}h^3 \\ \frac{2}{3}h^4 & \frac{h^4}{24} & 0 & \frac{h^4}{24} & \frac{2}{3}h^4 \end{pmatrix}.$$

and the coefficients $\mathbf{c} = (c_{-2}, c_{-1}, \ldots, c_2)^\top$ solve $\mathbf{Vc} = \mathbf{e}_3$ ($\mathbf{e}_3 = (0, 0, 1, 0, 0)^\top$).

2. Insert and compute...

3. Here's a matlab script creating the plot: which produces the plot

Figure 1: Convergence of difference approximation, Problem 4

and the shell output

```
h               D2u
1.0000e-01       6.4431e-05
5.6234e-02       6.4588e-06
3.1623e-02       6.4638e-07
1.7783e-02       6.4655e-08
1.0000e-02       6.4670e-09
5.6234e-03       6.4623e-10
3.1623e-03       6.4150e-11
1.7783e-03      -1.2502e-10
1.0000e-03      -4.1606e-10
5.6234e-04      -1.8130e-09
3.1623e-04       4.2406e-09
1.7783e-04       6.1032e-09
1.0000e-04      -3.8600e-08
```

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